

FuelCheck Data Retrieval and Processing

COMP5339 Project Assignment Stage 1

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Abstract

Stage 1 of the COMP5339 FuelCheck Data Retrieval and Processing project prepares real-world fuel pricing data for analysis and mapping. Web scraping, API access, structured integration, and preparation of over one million records was done. Timestamp decomposition, text field normalization, and member-only station flags and time-based transformations were key data preprocessing. Data exploration was performed to understand the data for Stage 2 Project. To enable geographical visualisation in the following project stage, we added geographic coordinates using an Australian Digital Atlas secondary dataset. A relational schema was designed in DuckDB with separate tables for static station details and dynamic fuel prices to ensure query efficiency and scalability. These laid the foundation for Stage 2 Project including mapping and visualisation.

1. Dataset Description

The dataset used in this project is sourced from the [NSW Government's FuelCheck portal](#), which provides latest fuel price data across petrol stations in New South Wales, Australia. This dataset provides monthly fuel price data in .csv and .xlsx formats. The dataset includes 8 features: station_name, address, suburb, postcode, brand, fuel_type, last_updated, and price.

a. Data Retrieval

A web scraping approach was used to automatically retrieve files for the years 2024 and 2025 by parsing HTML content using requests and BeautifulSoup. Regular expressions were used to filter file URLs based on year (2024, 2025, 24, or 25) and file format (CSV or XLSX), following which each dataset was downloaded and saved into a structured folder (fuelcheck_data/).

b. Data Integration

Using the pandas library, each file from the website was read into a DataFrame and append to a list. These DataFrames were then merged into a single DataFrame using pd.concat(all_dfs, ignore_index=True), which was then saved to a CSV file for further analysis. Files with corrupted formats or download issues were skipped with error handling in place to ensure the pipeline remained uninterrupted. The final file resulted in 8 features and 1,128,302 records.

c. Data Preprocessing

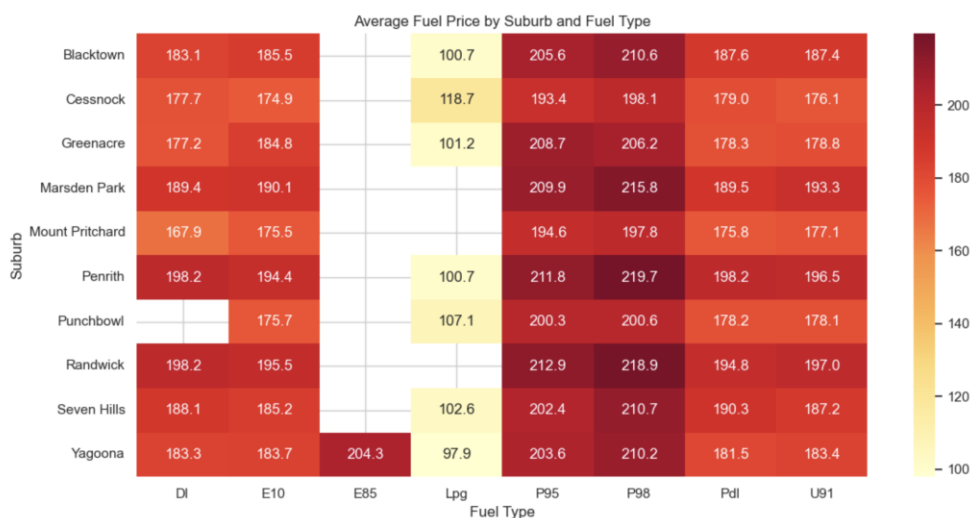
The following preprocessing steps were used for our dataset:

1. All column names were converted to lowercase, and spaces were replaced with underscores for consistency in column names.
2. Extra whitespace in text fields was stripped and also converted to title case.
3. The price column was converted to numeric (float) using pd.to_numeric(), and last_updated was converted to datetime format using pd.to_datetime().
4. Rows with null values, if any, were dropped from the dataset.
5. Duplicate rows, if any, were removed using drop_duplicates() to maintain data integrity.
6. To support time-based visualizations in Stage 2, the last_updated column was split into separate components: last_updated_year, last_updated_month, last_updated_day, and last_updated_time, and the original last_updated column was removed to avoid redundancy.
7. We created a separate column, members_only, to distinguish between public and members-only stations, indicating members only with true or false values.
8. The tag '(Members Only)' was removed from station_name values.

2. Data Exploration

Data exploration was carried out using pandas and matplotlib through various bar plots, box plots, line plot, heatmaps and scattered bar plots.

- **General exploration** was conducted by visualizing the average price by brand, the price distribution across suburbs, and the price per fuel type. Key findings included Costco having the lowest average fuel price, while Coles Express, BP, and Ampol are the most expensive. Marsden Park and Mount Pritchard show lower median prices, likely due to more Costco stations or less brand variation. High variance and medians in Greenacre and Casarock show a broad and competitive market. E85 and P98 are consistently the most expensive fuel types, while LPG is the cheapest. Outliers suggest price fluctuations depending on fuel type and location.
- **Time-based visualization** was done by plotting fuel prices across the months, average prices by day and month, and monthly average fuel price by brand and fuel type. It was found that April (month 4) had the highest average fuel prices, while September and October had the lowest. Some brands (like Coles Express and Ampol) have clear spikes mid-year, especially for premium fuels like P98, while Costco remains low for all fuel types. Diesel and P98 remained high no matter the brand.
- For Stage 2 mapping, visualization of brand distribution per suburb and the average fuel prices by suburb and fuel type were performed. Randwick and Seven Hills have the most variety of brands; Marsden Park has less variety, with Costco dominating, and brands like 7-Eleven and Ampol are consistent across suburbs. Diesel and E10 show price variation across suburbs. Fuel prices are generally lower in Mount Pritchard and Cessnock, indicating these areas are more affordable.
- Members-only stations show lower prices with less variance, while public stations have higher fuel prices, with the outliers indicating variance possibly due to location or fuel type.
- The following heatmap summarizes our key findings.



(Appendix [1])

These findings will help in using the dataset for layout, filters, color schemes, and data layers to create an interactive, informative, and user-focused map in project Stage 2.

3. Data Augmentation

Data augmentation involved integrated an addition external dataset from the Australian Digital Atlas, to enhance the given dataset for mapping and visualisation capabilities of our analysis. We augmented the given dataset (FuelCheck price data) with the geographic coordinates to accurately visualise it on the map.

The external dataset (Petrol_Station) contained X (longitude) and Y (latitude) coordinates of the overall fuel stations in Australia. The reason for choosing this dataset was its compatibility with the primary dataset,

these two shared attributes such as station name, station address, brand, postcode, suburbs. We mapped the coordinates of the fuel stations by using these common identifiers.

We created a unique station_key by concatenating cleaned and standardized version of station name, suburb and postcode. This key enabled joins between datasets by handling missing values and inconsistencies through normalization. After alignment, we joined the coordinate data to the corresponding fuel stations in the primary datasets.

4. Database Design

To normalise and structure the data for efficient querying and scalability, we used DuckDB to design and build relational database. We split the schema into two main tables: stations and fuel_prices. The stations table stores static information such as station ID (primary key), name, address, suburb, postcode, brand, and coordinates (longitude and latitude). The fuel_prices table stores dynamic attributes like station ID (foreign key), fuel type, price, members-only flag, and components of the last updated timestamp (year, month, day, time). This Schema helps in keeping the data organised, reduces repeating the same information, and supports analytical queries. We have designed it in a way that it is connected using join function, can breakdown to subgroup for better analysis like checking the prices of the E10 on 20th March 2025, and added longitude and latitude which later can be used for visualising for analytics in Stage 2.

a. Stations Table

This table stores information about petrol station.

- A unique ID for each station
- Station name (e.g., "Ampol Parramatta")
- Address, suburb, postcode
- Brand (like BP, Costco, etc.)
- The exact location using **latitude and longitude**

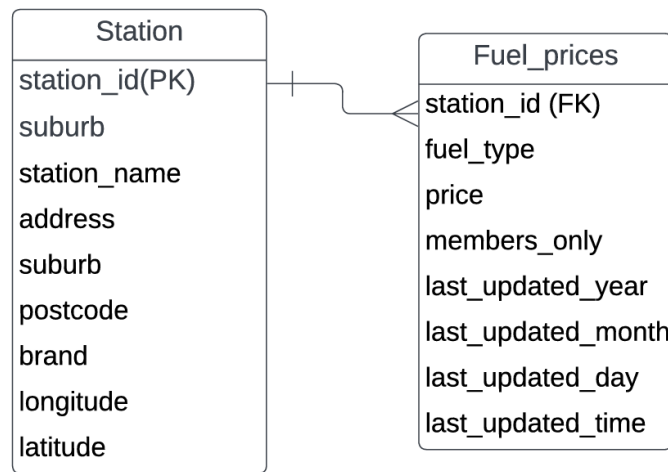
This info usually stays the same, so we only store it once per station.

b. Fuel Prices Table

This table stores the **changing** fuel price information.

- Which station the price belongs to (linked using the station ID)
- Type of fuel (like E10, Diesel, P98)
- The fuel price
- Whether it's a **members-only** station or not
- The exact **date and time** when the price was updated.

Our database schema can structured as follows:



(Appendix [2])

These two tables are connected by the station ID, so we can mix the information when needed. For example, if we want to know the price of E10 at Ampol in Parramatta on March 20th, 2025, we can get that answer by linking both tables.

Conclusion

We established a strong data pipeline to gather, process, and prepare NSW fuel pricing data for analysis in Stage 1 of COMP5339 FuelCheck. Over 1 million entries from several months and formats were merged using automated web scraping and API access. Thorough preprocessing maintained data integrity and reliability, enabling visual analysis and exploration. We identified key trends in pricing by brand, suburb, fuel type, and time, discovering valuable insights such as Costco's consistently lower prices and seasonal fluctuations in premium fuel types, along with the distribution of fuel brands across NSW suburbs.

To support geographic visualization in Stage 2, we augmented the dataset with geographical coordinates from a secondary source and standardized the data into a DuckDB schema for efficient searching and scalability. The structured design and improved dataset provide a solid foundation for developing an interactive, user-focused fuel price mapping application in the next phase. Stage 2 may easily visualize and explore spatial, brand, time, and fuel-type-based analysis thanks to this foundation.

Contribution

Adnan Ali: Contributed to data preprocessing and transformation tasks including cleaning, formatting, and extracting time-related fields. Worked on data transformation and storage pipelines.

Akarsh Kumar: Supported data augmentation by sourcing additional mapping datasets. Regularly participated in group discussions. Helped in drafting sections of the report and requirements.txt.

Akshat Jain: Contributed to data augmentation and helped define the file structure for organised storage. Focused on requirements.txt file. Took part majorly in team meetings and technical planning.

Annie Shorya: Focused on data preprocessing. Led the data exploration and insight generation phase. Played a key role in writing and structuring the final report and contributed throughout the team discussions.

Kevin Ninan Mathew: Led the data retrieval and integration phase. Also helped draft sections of the report and provided feedback in collaborative discussions.

Appendix

[1] The rows in the heatmap represent different suburbs in NSW, while the columns show different fuel types (DI, E10, E85, LPG, P95, P98, PDL, U91). Cell values show the average price (in cents) for each fuel type within each suburb. Darker red indicate higher prices and yellow/light orange represent lower prices.

[2] The left-side table (Station) contains static information about each fuel station, including station ID (primary key), name, address, suburb, postcode, brand, and geographical coordinates (latitude and longitude).

The right-side table (Fuel_prices) stores dynamic and time-based attributes such as fuel type, price, whether the station is members-only, and the date and time the price was last updated.

The two tables are connected via a one-to-many relationship, where each record in Station can have multiple corresponding records in Fuel_prices, linked through the station_id foreign key.
